ALLIANCE UNIVERSITY

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ALLIANCE UNIVERSITY

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Project Name: Airbnb Price Prediction and Market Opportunity Detection Using Machine Learning

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Signature of Student

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Signature of Faculty & Project Guide

1. Abstract:

Airbnb Price Insights is a data science and machine learning project developed to study Airbnb listing prices in Bengaluru and provide an estimated fair price for a listing based on its important characteristics. The project uses historical Airbnb listing data containing property, location, room type, capacity and price-related information. The main objective is not only to predict price, but also to provide a simple price insight by comparing the actual listing price with the model's estimated price.

The project follows a practical data science workflow. The dataset is first inspected to understand its structure, data types, missing values, duplicates and numerical distributions. Missing values are handled using mode for selected categorical variables and median for selected numerical variables. Outlier analysis is performed using the Interquartile Range (IQR) method. Extremely large property-size values are reviewed rather than removing every statistical outlier. A logarithmic transformation is applied to the price because the original price distribution is strongly right-skewed.

For prediction, the selected input features are RoomType, City, Bedrooms, Bathrooms, Beds and PersonCapacity. Categorical variables are converted into numerical form using One-Hot Encoding. A Linear Regression model is trained on the transformed data. The model is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared (R^2). The final model produced an MAE of approximately ₹1,637.59, an RMSE of approximately ₹3,424.27 and an R^2 value of approximately 0.3035 on the test data.

A simple opportunity detection component compares actual price with predicted price. A difference greater than ₹500 is classified as possibly overpriced, a difference below -₹500 is classified as possibly underpriced, and values within this range are classified as fair price. The word 'possibly' is used because the result is an estimate based on the trained model and should not be treated as a definitive market valuation.

2. Introduction:

Short-term rental platforms contain a large amount of information about accommodation properties. A listing can differ from another listing in room type, number of bedrooms, bathrooms, beds, capacity and location. These characteristics can influence the price that a host charges. For a user or property owner, simply looking at the listed price does not always show whether the price is reasonable compared with similar properties.

Data science provides a way to study these patterns using historical information. By analysing previous listings, a machine learning model can learn relationships between property characteristics and price. The resulting model can then be used to estimate the price of a new listing when the relevant characteristics are supplied.

The Airbnb Price Insights project was developed with this idea. The project uses a Bengaluru Airbnb dataset and focuses on a small set of understandable features. The aim is to keep the analysis appropriate for an academic machine learning project while still building a complete workflow from data analysis to a working prediction interface.

The project is divided into two major parts. The first part is the data science and machine learning workflow performed in a Jupyter Notebook. This includes data loading, exploratory analysis, missing-value treatment, outlier analysis, transformation, feature selection, model training and evaluation. The second part is the application layer. A Flask-based ML API loads the trained model and preprocessor and receives listing details from a simple HTML, CSS and JavaScript frontend.

2.1 Objectives

- To understand and analyse Airbnb listing data for Bengaluru.
- To identify important property and location characteristics related to listing price.
- To clean and prepare the dataset for machine learning.
- To handle missing values and investigate extreme observations.
- To reduce the effect of strong price skewness through logarithmic transformation.
- To build a simple Linear Regression model for price estimation.
- To evaluate the prediction model using standard regression metrics.
- To compare an actual listing price with the estimated fair price.

- To provide a simple web-based interface for prediction.

2.2 Scope

The scope of the project is limited to price estimation using selected listing characteristics, including room type, city, number of bedrooms, bathrooms, beds, and person capacity. These features are used by the machine learning model to estimate the expected nightly price of an Airbnb listing. The project focuses on understanding how these property characteristics relate to price and how historical listing data can be used to generate useful price estimates.

The system also includes a market opportunity detection component that compares the actual listing price with the predicted price. Based on the difference, a listing is classified as possibly underpriced, fairly priced, or possibly overpriced. This feature helps users make an initial comparison between a property's current price and the price estimated by the model. However, these classifications are indicative only and should not be considered a guarantee of a property's actual market value or investment potential.

The project does not attempt to provide a complete business valuation, guarantee profitability, recommend property investments, or determine the exact market price of a property. Airbnb prices may also depend on factors such as seasonal demand, customer reviews, amenities, property condition, neighbourhood popularity, availability, and local events, which are not fully considered by the selected model. Therefore, the predictions may differ from actual listing prices, and the results should be interpreted with these limitations in mind.

From a technical perspective, the project covers the major stages of a basic machine learning workflow, including data inspection, data cleaning, missing-value treatment, outlier analysis, exploratory data analysis, feature selection, data preprocessing, model training, and performance evaluation. The trained model is then connected to a simple web application using Flask, HTML, CSS, and JavaScript, allowing users to enter property details and view the estimated price without directly interacting with the notebook.

Overall, the project is intended as an educational demonstration of data analysis, machine learning, regression-based price prediction, and basic application integration. Its primary purpose is to show how historical data can be transformed into useful information through a structured analytical process and presented through an interactive interface. The project provides a foundation for further improvements, such as including additional listing features, comparing different regression algorithms, and using more recent data to improve prediction performance.

3. Problem Statement:

Airbnb listings can have significantly different prices even when they are located within the same broad city area. The price of a property may vary depending on several characteristics, such as room type, number of bedrooms, bathrooms, beds, guest capacity, and location. For example, an entire home or flat may have a different price from a private room or shared room because of differences in space, privacy, and accommodation facilities. Similarly, properties with more bedrooms, bathrooms, or higher guest capacity may have different pricing requirements.

For a person evaluating an Airbnb listing, it can be difficult to decide whether the displayed nightly price is reasonable by looking at the price alone. Manual comparison of multiple listings can be time-consuming, particularly when the properties have different characteristics. A simple average price may also be insufficient because it does not consider the individual features of a particular property. Therefore, a systematic approach is required to analyse historical listing data and estimate a suitable price based on the characteristics of a property.

The main problem addressed by this project is to develop a machine learning system that can estimate the expected nightly price of an Airbnb listing using its basic characteristics. The system uses historical Bengaluru Airbnb listing data to identify patterns between selected property features and prices. A Linear Regression model is trained to learn these relationships and generate price predictions for listings with specified input values. The predicted price is then converted back into the original currency scale so that users can understand the result in Indian rupees.

In addition to price prediction, the project aims to provide a simple market opportunity indicator. The application compares the actual nightly price entered by the user with the estimated price generated by the model. Based on the difference, the listing is categorised as possibly underpriced, fairly priced, or possibly overpriced. This comparison provides an additional way to interpret the prediction and helps users understand how the actual price differs from the model's estimate. However, these categories represent preliminary indicators and do not guarantee the actual market value of a listing.

Another important problem is the quality and consistency of the dataset used for analysis. Real-world datasets may contain missing values, unusual observations, and numerical values that differ considerably from the majority of records. If these issues are not investigated, they can affect descriptive statistics, visualisations, and model performance. The project therefore includes data inspection, missing-value treatment, exploratory data analysis, outlier investigation, and logarithmic transformation of the price variable before model training.

The project also addresses the need to make machine learning results accessible through a simple interface. Instead of requiring users to run notebook cells or understand the underlying Python code, the trained model is connected to a web application using Flask, HTML, CSS, and JavaScript. Users can enter the relevant property details, request a prediction, and view the estimated price and price comparison through the interface.

Overall, the problem is to create an understandable and practical system that combines data analysis, machine learning-based price estimation, and basic market opportunity detection. The system is intended to demonstrate how historical data can be processed and used to support price comparison, while recognising that the accuracy of predictions depends on the available data, the selected features, and the performance of the trained model.

3.1 Key Questions

1. **Can listing characteristics be used to estimate price?**
   Can features such as room type, city, number of bedrooms, bathrooms, beds, and person capacity help a machine learning model estimate the nightly price of an Airbnb listing? The project examines these features to understand their relationship with price and their usefulness for prediction.
2. **How does the original price distribution affect the modelling process?**
   Does the presence of very high-priced listings create a strongly right-skewed price distribution, and how can this affect the regression model? The project investigates the distribution using descriptive statistics, a histogram, and a boxplot. A logarithmic transformation is then applied to the target price to reduce the effect of strong skewness during modelling.
3. **How can missing values be handled using simple statistical methods?**
   How can incomplete records be prepared for analysis without unnecessarily discarding useful data? The project uses the mode to fill selected categorical values and the median to fill selected numerical values. These methods provide a straightforward way to handle missing values in the selected columns.
4. **Which evaluation metrics can be used to understand the performance of the regression model?**
   How closely do the predicted prices match the actual prices in the test dataset? The project uses Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared (R^2) score. These metrics help measure prediction errors and explain how much variation in price is accounted for by the model.
5. **Can the difference between actual and predicted price be used to create a simple price opportunity indicator?**
   Can the difference between the actual listing price and the model's estimated price provide a useful preliminary comparison? The project applies a ₹500 threshold to classify listings as possibly underpriced, fairly priced, or possibly overpriced. This approach makes the model output easier to interpret while acknowledging that the classifications are estimates rather than confirmed market judgements.

4. Existing System:

In a basic manual approach, a user may compare Airbnb listings by looking at displayed prices and a few visible property details. This approach depends heavily on individual judgement. Two properties may appear similar, but their prices may differ because of multiple characteristics that are difficult to compare consistently.

Another common approach is to use the average or median price of a group of listings. While such summaries are useful for understanding the market, they do not consider the characteristics of an individual property. A small private room, for example, should not necessarily be compared directly with a large entire home.

The existing manual approach therefore has several limitations. It may be time-consuming, it may not provide a consistent comparison, and it does not automatically provide a model-based estimated price for a new combination of property features.

4.1 Limitations

- **Price Comparison Is Largely Manual:** Users often compare Airbnb prices by checking individual listings, which can be time-consuming. Manual comparisons may also lead to inconsistent results.
- **Difficulty in Comparing Properties with Different Characteristics:** Properties differ in room type, bedrooms, bathrooms, beds, capacity, and location. Comparing prices without considering these differences may lead to misleading conclusions.
- **Limitations of Using a Simple Average Price:** An average price provides a general overview but does not consider individual property features. Expensive listings may also increase the average and make it less representative.
- **Absence of Automatic Price Prediction:** Traditional methods do not automatically estimate the price of a new listing based on its characteristics. Users must rely on manual comparisons or general price information.

  These limitations highlight the need for a systematic, data-driven approach. The proposed system uses historical Airbnb data and machine learning to estimate nightly prices and compare them with actual listing prices, providing users with a more structured way to understand potential price differences.

5. Proposed System:

The proposed Airbnb Price Insights system combines data analysis, machine learning and a simple web interface. The system takes selected listing details as input and sends them to a trained Linear Regression model through a Flask API. The model returns an estimated price. If an actual listing price is also supplied, the application calculates the difference between the actual and estimated price and displays a simple classification.

The proposed system begins with historical data preparation. The dataset is inspected for its shape, data types, missing values, unique values and duplicates. Selected missing values are filled using the mode for categorical variables and the median for numerical variables. Potential outliers are investigated using IQR. Extremely large values in Bedrooms, Bathrooms and Beds are removed from the modelling copy of the data when they exceed the selected practical limit of 10.

The price target is strongly right-skewed. Therefore, the project creates LogPrice using log1p(price). This transformed target is used during model training. After prediction, the logarithmic value is converted back into the original rupee scale using the inverse exponential transformation.

The selected features are RoomType, City, Bedrooms, Bathrooms, Beds and PersonCapacity. RoomType and City are categorical variables and are encoded using One-Hot Encoding. Numerical variables are retained as numerical inputs. The data is split into training and testing sets using an 80:20 split with random_state=42. The preprocessor is fitted only on the training data and then applied to the test data to avoid information leakage.

5.1 Proposed System Workflow

6. Load the Bengaluru Airbnb dataset.
7. Inspect the structure, data types, missing values, duplicates and distributions.
8. Handle selected missing values using mode and median.
9. Investigate price and property-size outliers using IQR and practical limits.
10. Create a modelling copy and apply logarithmic transformation to price.
11. Select the six input features used by the model.
12. Split the data into training and testing sets.

13. Apply One-Hot Encoding to categorical variables.

14. Train a Linear Regression model.

15. Predict the transformed price and convert it back to rupee price.

16. Evaluate the model using MAE, MSE, RMSE and R^2.

17. Compare actual and predicted prices for opportunity detection.

18. Save the model and preprocessor for use by the Flask API.

19. Connect the web frontend to the prediction API.

5.2 Opportunity Detection Logic

The Opportunity Detection component compares the actual nightly price of an Airbnb listing with the price predicted by the machine learning model. It calculates the difference using the following formula:

Price Difference = Actual Price − Predicted Price

Based on the price difference, the application classifies listings into three categories:

1. Possibly Overpriced: If the actual price is more than ₹500 above the predicted price, the listing is labelled "Possibly Overpriced." For example, if the predicted price is ₹3,000 and the actual price is ₹3,800, the difference is ₹800.

2. Possibly Underpriced: If the actual price is more than ₹500 below the predicted price, the listing is labelled "Possibly Underpriced." For example, if the predicted price is ₹3,500 and the actual price is ₹2,700, the difference is −₹800.

3. Fair Price: If the difference falls within the range of −₹500 to +₹500, the listing is labelled "Fair Price." This indicates that the actual price is relatively close to the predicted price.

The ₹500 threshold is a simple project-level rule, not a statistical confidence interval. These categories provide an initial indication of pricing differences and do not confirm whether a listing is genuinely overpriced or underpriced.

Through this approach, the application combines machine learning-based price prediction with a straightforward rule-based classification system. It presents the predicted price, compares it with the actual price, calculates the difference, and displays the corresponding category. This makes the results easier for users to understand and demonstrates how data analysis and machine learning can support preliminary price evaluation in the short-term rental market.

6. About Dataset:

The project uses a Bengaluru Airbnb listing dataset containing 6,002 records and 68 columns in the original dataset. The dataset contains information describing Airbnb properties, including room type, city, bedrooms, bathrooms, beds, person capacity and the basic night price used as the target variable.

The initial analysis showed a mixture of numerical and categorical information and several columns with missing values. The RoomType variable contained three major categories: Entire home/flat, Private room and Shared room. Entire home/flat was the largest category in the dataset.

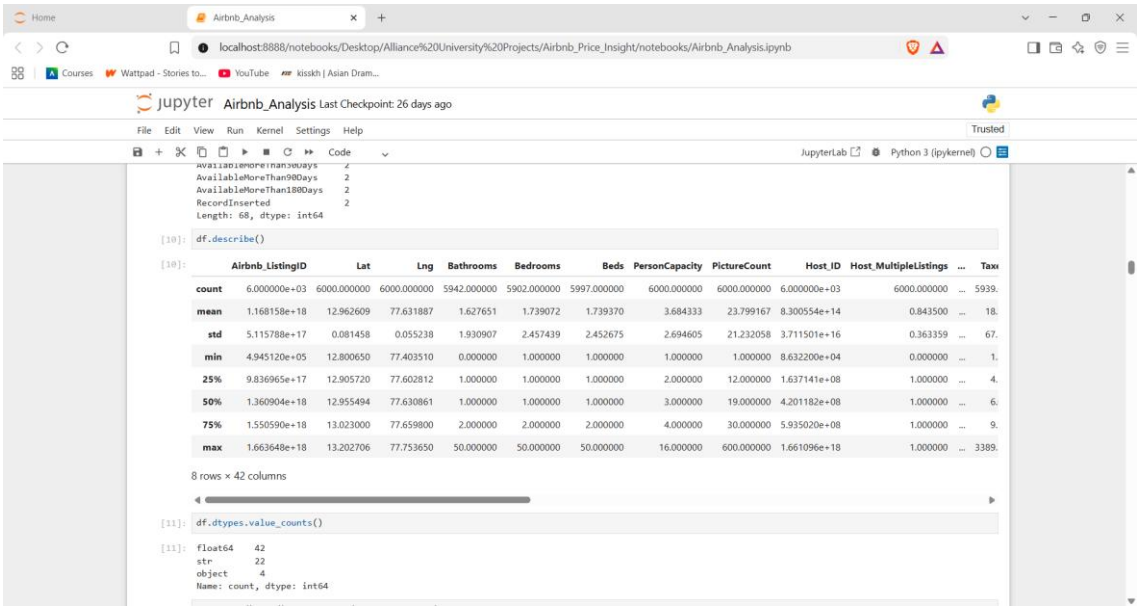
The BasicNightPrice variable ranged from approximately ₹438 to ₹350,400. Its mean was approximately ₹4,093.64 while its median was approximately ₹2,598.50. The difference between the mean and median and the long upper tail in the distribution indicated strong right skewness.

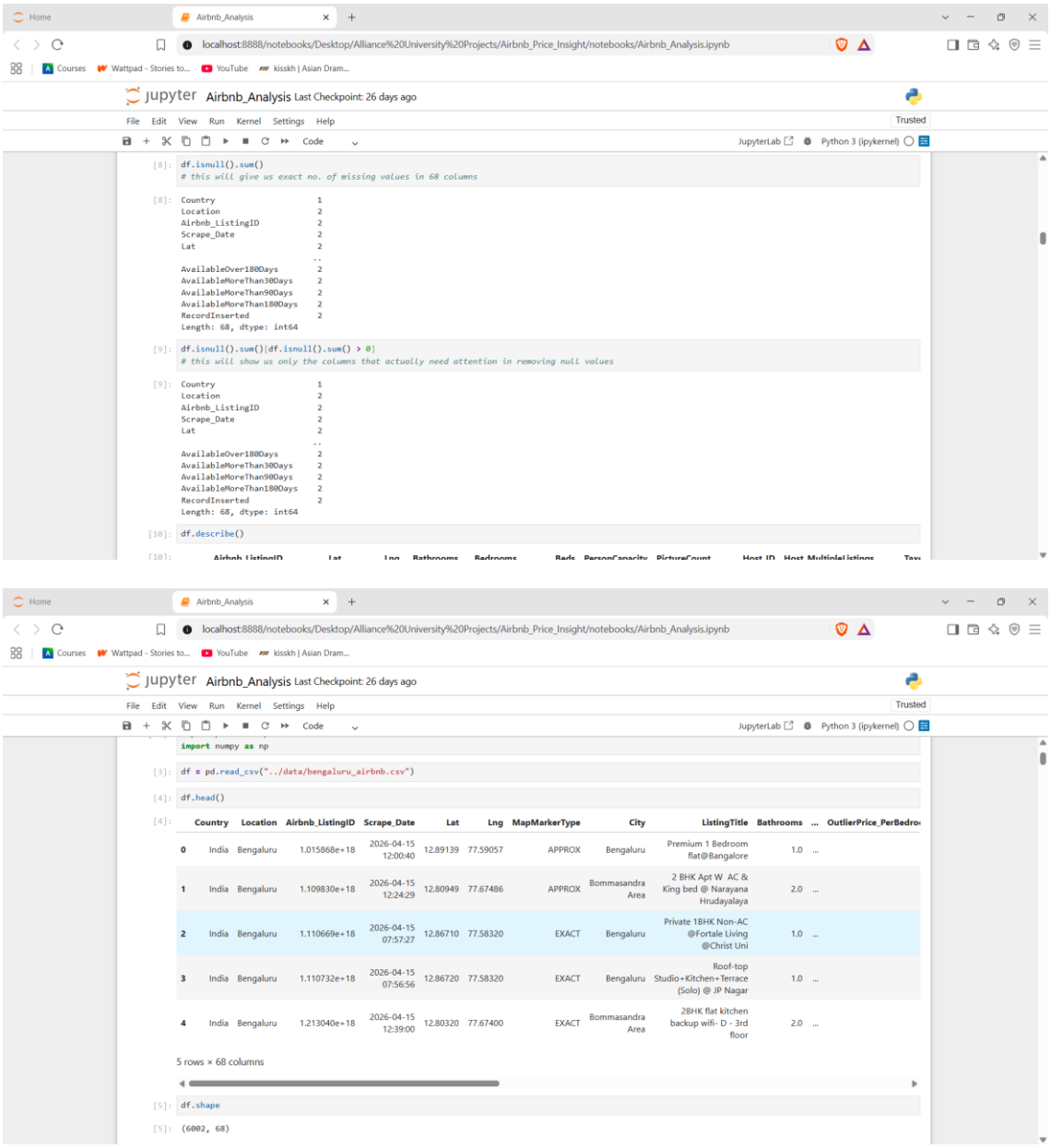
6.1 Data Cleaning

Figure 6.1: Dataset shape and initial dataset inspection.

Figure 6.2: Initial missing-value analysis.

Figure 6.3: Descriptive statistical summary of numerical variables.

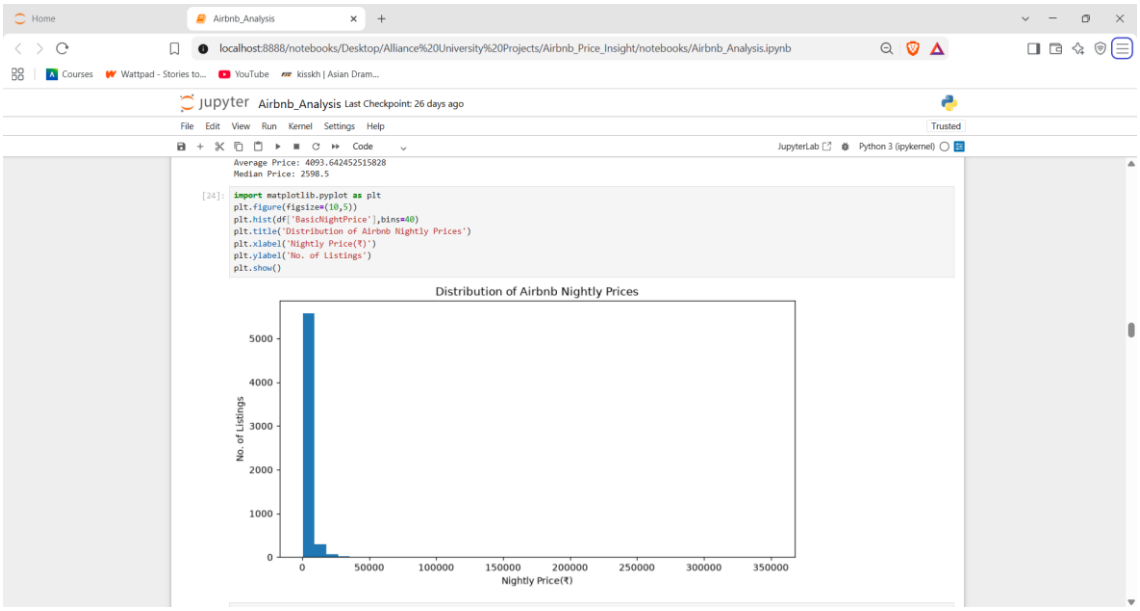




Duplicate-row analysis returned zero duplicate rows. Selected missing categorical values were filled using the mode, while selected numerical values were filled using the median. The selected variables were RoomType, City, Bedrooms, Bathrooms, PersonCapacity and BasicNightPrice.

6.2 Outlier Analysis

Figure 6.4: Distribution of Airbnb nightly prices.



For BasicNightPrice, the first quartile was ₹1,776 and the third quartile was ₹4,064. Therefore, IQR was ₹2,288. Using the 1.5 × IQR rule produced a lower fence of approximately -₹1,656 and an upper fence of approximately ₹7,496. About 590 observations were identified as statistical price outliers. These observations were not automatically deleted because a high Airbnb price can represent a genuine high-value property.

Additional practical checks were performed for property-size variables. There were 50 records with Bedrooms greater than 10, 30 records with Bathrooms greater than 10 and 51 records with Beds greater than 10. PersonCapacity had no values above 16. For the modelling copy, listings with Bedrooms, Bathrooms or Beds above 10 were excluded, leaving 5,941 records for modelling. The original dataframe was retained separately.

6.3 Feature Selection

The final model uses six features: RoomType, City, Bedrooms, Bathrooms, Beds and PersonCapacity. BasicNightPrice itself was not used as an input feature because it is the target being predicted. Including the actual target price as an input would create target leakage and would not represent a real prediction scenario.

6.4 Exploratory Findings

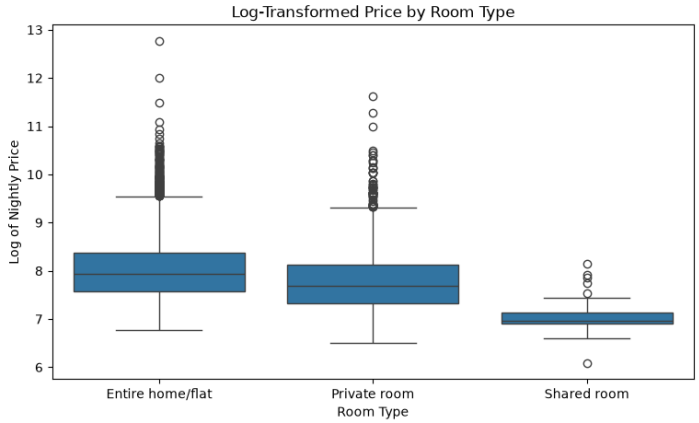
Figure 6.5: Log-transformed price comparison by room type.

```
[44]: plt.figure(figsize=(9, 5))

sns.boxplot(
    data=df_model,
    x='RoomType',
    y='LogPrice'
)

plt.title('Log-Transformed Price by Room Type')
plt.xlabel('Room Type')
plt.ylabel('Log of Nightly Price')

plt.show()
```



The mean price differed across room types. Entire home/flat listings had an average price of approximately ₹4,327.47, Private room listings had an average of approximately ₹3,264.62, and Shared room listings had an average of approximately ₹1,274.87. The Shared room group contained only 38 listings in the cleaned modelling data, so its average should be interpreted carefully.

The correlation analysis with LogPrice showed the strongest relationships among the selected numerical features for PersonCapacity (approximately 0.5902), Bathrooms (approximately 0.5509), Bedrooms (approximately 0.5470) and Beds (approximately 0.5334). These values supported the selection of property-size and capacity features for the model.

Item	Value	Description
Original records	6,002	Records in original dataset
Original columns	68	Variables available
Duplicate rows	0	No duplicate rows found
Modelling records	5,941	After practical property-size filtering
Target	BasicNightPrice	Nightly price in rupees
Transformation	log1p	Reduces strong right skewness
Input features	6	RoomType, City, Bedrooms, Bathrooms, Beds, PersonCapacity

7. Technologies Used:

The project uses a combination of data science, machine learning, Python programming and basic web technologies. The selected tools were chosen to keep the system understandable and suitable for the project requirements.

Technology / Tool	Purpose	Use in Project
Python	Programming	Analysis, preprocessing and machine learning
Pandas	Data manipulation	Loading, cleaning and analysing data
NumPy	Numerical computation	Mathematical operations and transformation
Matplotlib	Visualization	Histograms, boxplots, scatter and result charts
Scikit-learn	Machine learning	Encoding, Linear Regression and evaluation
Jupyter Notebook	Development	Step-by-step analysis and modelling
Flask	Backend API	Provides the /predict endpoint
HTML/CSS/JavaScript	Frontend	Collects inputs and displays results

7.1 Machine Learning Components

- Train-test split: 80% training and 20% testing.
- One-Hot Encoding: converts RoomType and City into numerical inputs.
- Linear Regression: primary regression algorithm.
- Log transformation: applied to target price.
- MAE, MSE, RMSE and R^2: model evaluation.
- Joblib: saves the trained model and preprocessor.

7.2 Application Components

The frontend contains an input form for Room Type, City, Bedrooms, Bathrooms, Beds, Person Capacity and optional Current Price. JavaScript sends these values as JSON to the Flask API. The API applies the saved preprocessor, makes the prediction, converts the transformed prediction back to rupees and returns the result as JSON. The frontend then displays the estimated price and, when an actual price is entered, the price difference and opportunity classification.

8. Data Flow Diagram:

The data flow of the proposed system can be represented as follows:

Stage	Input / Process	Output
1. Dataset	Bengaluru Airbnb CSV dataset	Raw listing data
2. Data Preparation	Missing values, duplicate check and outlier analysis	Clean modelling data
3. Transformation	Log price transformation and One-Hot Encoding	Model-ready data
4. Model Training	80:20 split and Linear Regression	Trained price model
5. Model Saving	Joblib saves model and preprocessor	PKL model files
6. User Prediction	Frontend sends listing details to Flask /predict	Predicted fair price

8.1 Text Representation of Data Flow

Historical Airbnb Dataset → Data Cleaning → Outlier Investigation → Price Transformation → Feature Encoding → Train/Test Split → Linear Regression Model → Model Evaluation → Saved Model.

User Input → HTML/CSS/JavaScript Frontend → Flask /predict API → Saved Preprocessor → Linear Regression Model → Predicted Price → Price Difference → Opportunity Result.

8.2 API Flow

20. The user enters listing information in the web form.
21. JavaScript collects the values and creates a JSON request.
22. The request is sent to the Flask /predict endpoint.
23. The saved preprocessor transforms the input into the same feature format used during training.
24. The saved Linear Regression model predicts LogPrice.
25. The API converts the prediction back to rupee price.
26. The prediction is returned to the frontend.
27. The frontend displays the result and compares it with actual price if supplied.

9. Output:

The project produces outputs at both notebook and application levels. In the notebook, the output includes descriptive statistics, missing-value checks, outlier analysis, room-type price summaries, correlation analysis, actual-versus-predicted comparisons and an opportunity detection summary.

9.1 Model Performance

Figure 9.1: Actual versus predicted Airbnb prices.

```
[61]: actual_price = np.exp(y_test.values[:30])

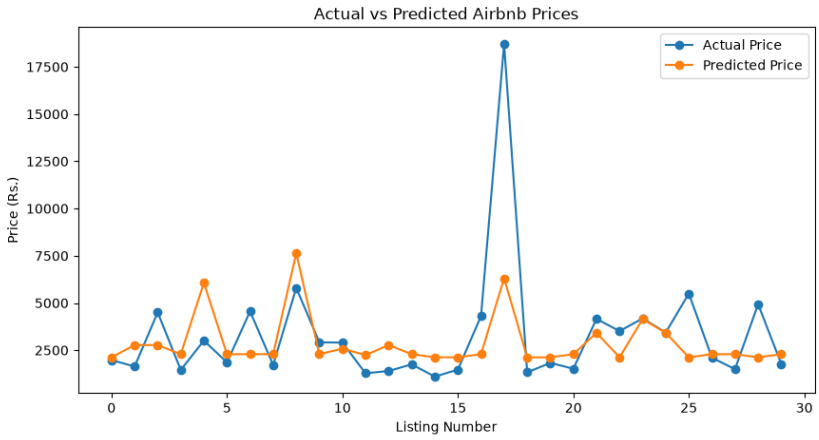
      predicted_price_30 = predicted_price[:30]

      plt.figure(figsize=(10, 5))

      plt.plot(actual_price, marker='o', label='Actual Price')
      plt.plot(predicted_price_30, marker='o', label='Predicted Price')

      plt.xlabel("Listing Number")
      plt.ylabel("Price (Rs.)")
      plt.title("Actual vs Predicted Airbnb Prices")

      plt.legend()
      plt.show()
```



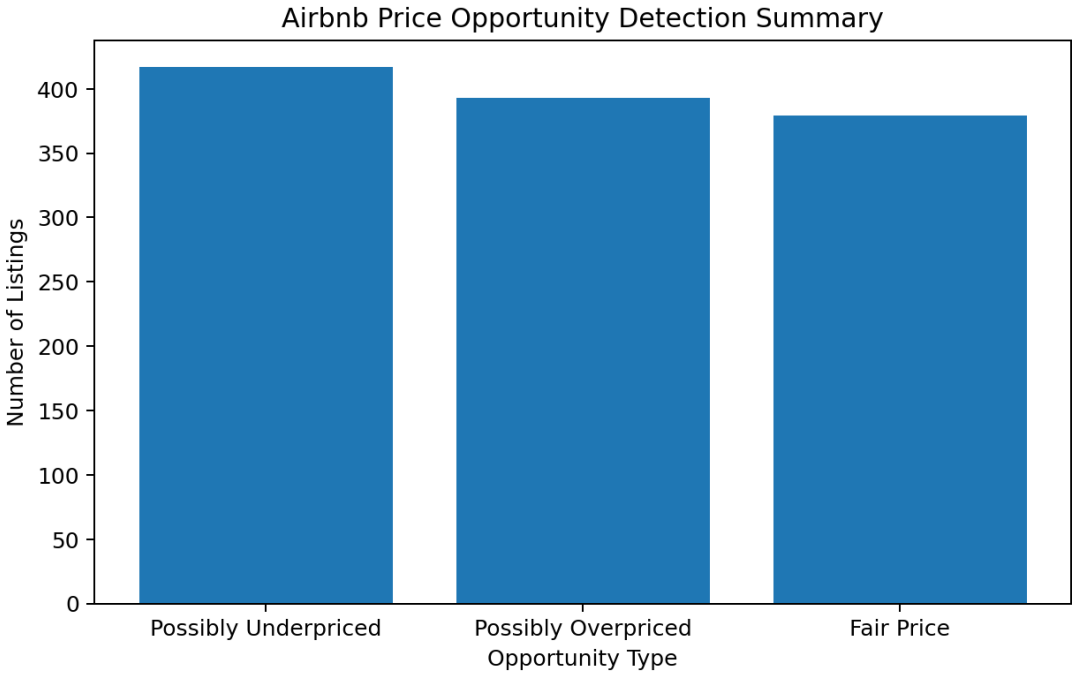
Metric	Result	Interpretation
MAE	₹1,637.59	Average absolute prediction difference
MSE	11,725,619.87	Mean squared prediction error
RMSE	₹3,424.27	Error scale with greater penalty for large errors
R^2	0.3035	About 30.35% of variation explained

The results show that the model provides a useful baseline for price estimation but does not explain all variation in Airbnb prices. An MAE of approximately ₹1,637.59 means that the average absolute difference between actual and predicted prices in the test set is around ₹1,638. The RMSE is higher because larger errors have a greater influence on this metric. The R^2 value

of 0.3035 indicates that the selected features explain around 30.35% of the variation represented by the evaluation.

9.2 Opportunity Detection Output

Figure 9.2: Opportunity detection summary based on test-set predictions.



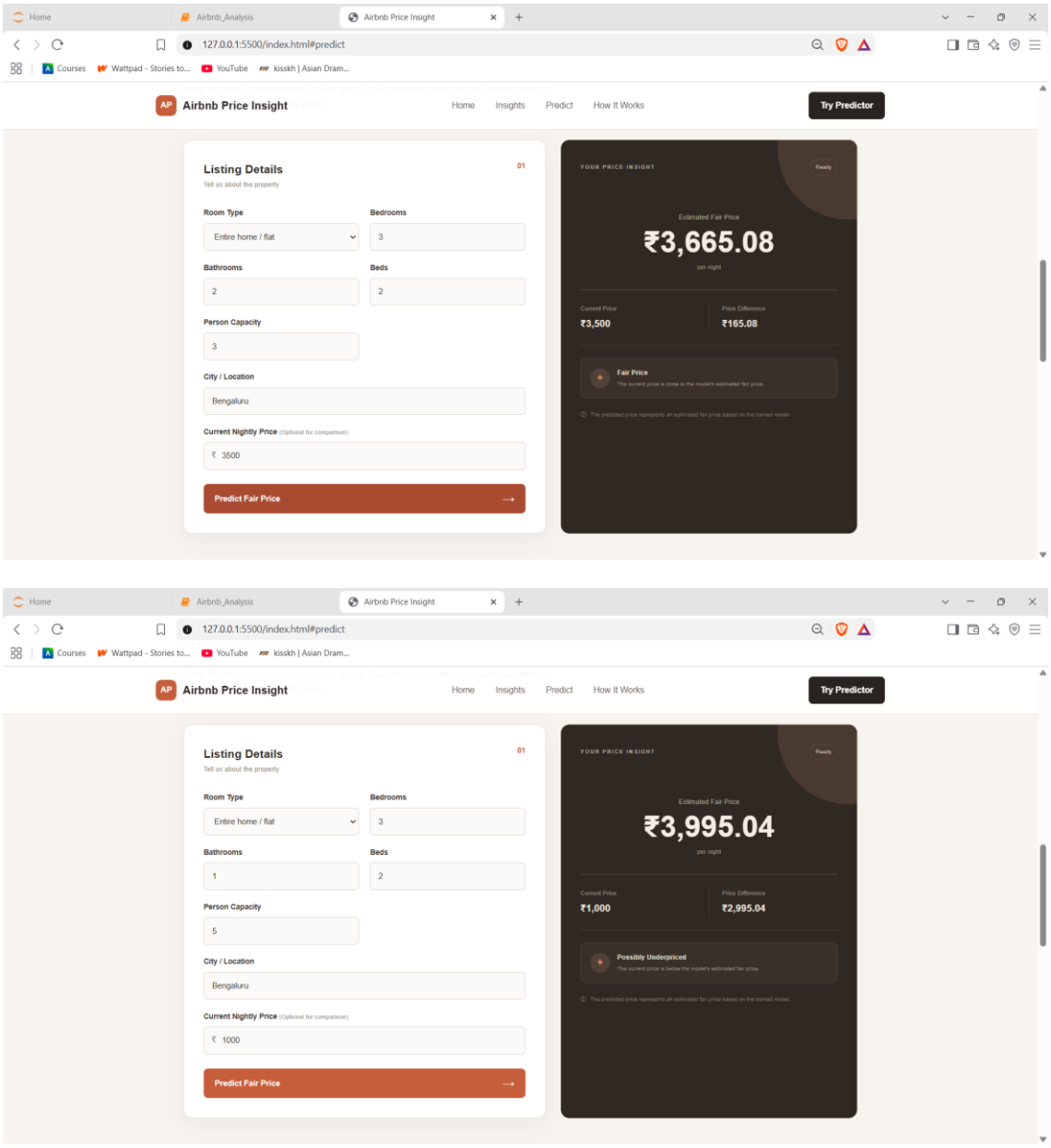
Category	Count	Meaning
Possibly Underpriced	417	Actual price is more than ₹500 below predicted
Possibly Overpriced	393	Actual price is more than ₹500 above predicted
Fair Price	379	Actual price is within ±₹500

The opportunity detection counts are based on the test-set predictions and the selected ₹500 threshold. They are model-based indicators rather than confirmed market judgements.

9.3 Example Application Output

Figure 9.3: Frontend prediction screen showing an estimated fair price and a possibly underpriced result.

Figure 9.4: Frontend screen showing a fair-price result.



For a test input with Room Type = Entire home/flat, City = Bengaluru, Bedrooms = 2, Bathrooms = 2, Beds = 2 and Person Capacity = 4, the application produced an estimated price of approximately ₹3,421.99.

When the current price was entered as ₹3,000, the difference from the prediction was approximately ₹421.99, which falls within the ±₹500 threshold and was therefore displayed as 'Fair Price'. When a much lower current price such as ₹300 was entered, the listing was classified as 'Possibly Underpriced'.

The frontend also supports the 'Possibly Overpriced' case when the actual price is more than ₹500 above the estimated price. This gives the user three simple and understandable price insight categories.

9.2 Screenshots

The following screenshots demonstrate the implementation and working of the Airbnb Price Prediction and Market Opportunity Detection Using Machine Learning project. They cover the major stages, from dataset analysis and price visualization to machine learning predictions and frontend results.

Figure 1: Dataset Overview

This screenshot shows the initial inspection of the Airbnb dataset, including its structure, number of records, columns, and sample data. It provides an overview of the information used for the project.

Figure 2: Data Cleaning and Preprocessing

This screenshot presents the initial data preparation process, including checking missing values and examining the dataset before analysis. These steps help prepare the data for further processing.

Figure 3: Exploratory Data Analysis

This screenshot shows the initial analysis of the dataset using Python. It helps understand the data and identify important characteristics of Airbnb listings.

Figure 4: Price Distribution Analysis

This screenshot presents the price distribution using a histogram and boxplot. The visualizations help identify the distribution of nightly prices and potential outliers in the dataset.

Figure 5: Room Type Price Comparison

This screenshot compares prices across different room types using log-transformed price values. It helps examine how pricing varies between entire homes or flats, private rooms, and shared rooms.

Figure 6: Actual vs. Predicted Price Comparison

This screenshot compares the actual prices with the prices predicted by the Linear Regression model. It provides a visual understanding of how closely the model's predictions match the actual values.

Figure 7: Model Prediction Analysis

This screenshot presents another visualization of actual and predicted prices. It provides an additional view of the model's performance and helps identify differences between actual prices and estimated prices.

Figure 8: Frontend Price Prediction

This screenshot shows the frontend interface where users can enter property details, such as room type, city, number of bedrooms, bathrooms, beds, and guest capacity. The application uses these details to generate an estimated nightly price.

Figure 9: Frontend Opportunity Detection

This screenshot demonstrates the opportunity detection feature, which compares the actual price with the predicted price. Based on the ₹500 threshold, the application classifies the listing as Possibly Underpriced, Possibly Overpriced, or Fair Price.

Overall, these screenshots provide visual evidence of the different stages involved in developing the Airbnb Price Prediction and Market Opportunity Detection Using Machine Learning project. They illustrate the process of inspecting and preparing the dataset, performing exploratory data analysis, examining price distributions, comparing prices across room types, and evaluating the results of the Linear Regression model. The frontend screenshots further demonstrate how users can enter property details, obtain an estimated nightly price, and compare it with the actual listing price. Based on the selected ₹500 threshold, the application classifies listings as Possibly Underpriced, Possibly Overpriced, or Fair Price. Together, these figures explain how the project integrates data analysis, machine learning, and a user-friendly interface into a single application. They also help readers understand how raw Airbnb listing data is converted into meaningful visualizations and preliminary pricing insights. Although the results can support an initial comparison of listing prices, the classifications are estimates and should not be considered a guarantee of a property's actual market value.

10. Conclusion:

The Airbnb Price Insights project demonstrates a complete beginner-to-intermediate data science workflow using a real-world style Airbnb listing dataset. The project begins with data understanding and cleaning and progresses through exploratory data analysis, outlier investigation, transformation, feature selection, machine learning, evaluation and application integration.

The analysis showed that Airbnb price is strongly skewed, with a relatively small number of high-priced listings. Instead of blindly deleting every statistical outlier, the project investigates the data and applies practical limits only to extreme property-size values used for modelling. The logarithmic transformation of price provides a more suitable target for the regression model.

A Linear Regression model was selected because it is simple, interpretable and appropriate for demonstrating the basic machine learning workflow taught in the project. The model achieved an MAE of approximately ₹1,637.59, RMSE of approximately ₹3,424.27 and R^2 of approximately 0.3035. These results indicate that the model provides a useful baseline while also showing that Airbnb pricing depends on factors beyond the six selected features.

The opportunity detection component adds practical value by comparing the actual listing price with the estimated price. The categories 'Possibly Underpriced', 'Fair Price' and 'Possibly Overpriced' make the model output easier for a user to understand. However, these categories are indicators generated by the model and should not be treated as guaranteed market judgements.

Finally, the project connects the machine learning model to a simple web interface using Flask, HTML, CSS and JavaScript. This demonstrates how a trained model can move beyond a notebook and become part of an interactive application. The project therefore provides both an analytical component and a practical demonstration component.

10.1 Limitations

- The model uses a limited set of listing features and cannot capture every factor affecting price.
- The R^2 value shows that substantial price variation remains unexplained.
- The dataset represents a particular set of Airbnb listings and may not represent all future market conditions.

- The ₹500 opportunity threshold is a simple project rule and not a formal statistical confidence boundary.
- The system provides an estimated price and should not be treated as a guaranteed market valuation.

10.2 Future Scope

- Additional relevant listing and location features can be included after proper validation.
- Larger and more recent datasets can be used to improve generalisation.
- More regression algorithms can be compared as an extension.
- A deployed cloud version of the frontend and API can be created.
- A dashboard can be added for broader market-level analysis.
- More robust statistical methods can be used to estimate prediction uncertainty.

References

- **Bengaluru Airbnb Listing Dataset:** Dataset used for data cleaning, exploratory data analysis, price prediction, and market opportunity detection.
- **Python Documentation:** Official documentation for Python programming concepts and functions used in the project.
- **Pandas and NumPy Documentation:** References for data manipulation, handling missing values, numerical calculations, and statistical analysis.
- **Scikit-learn Documentation:** Reference for data preprocessing, One-Hot Encoding, Linear Regression, train-test splitting, and model evaluation.
- **Flask Documentation:** Reference for developing the prediction API that connects the trained machine learning model with the frontend application.
- **Joblib Documentation:** Reference for saving and loading the trained machine learning model and preprocessing objects.
- **HTML, CSS, and JavaScript Documentation:** References for developing the frontend interface, styling the application, collecting user inputs, and displaying prediction results.